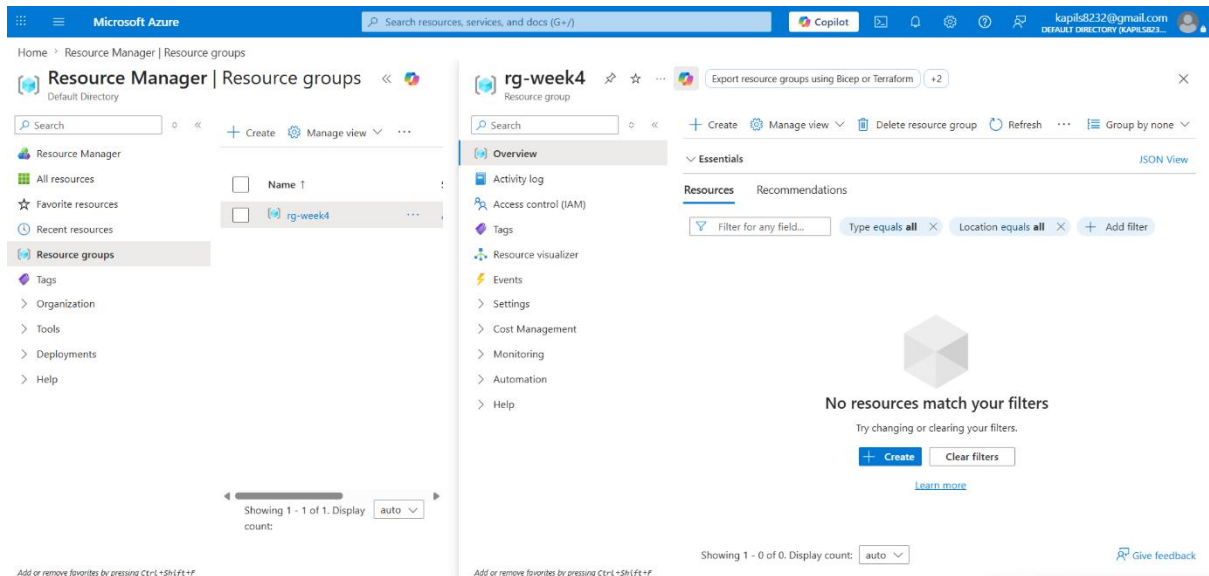


Objective: Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory.

Step 1: Resource Group Creation

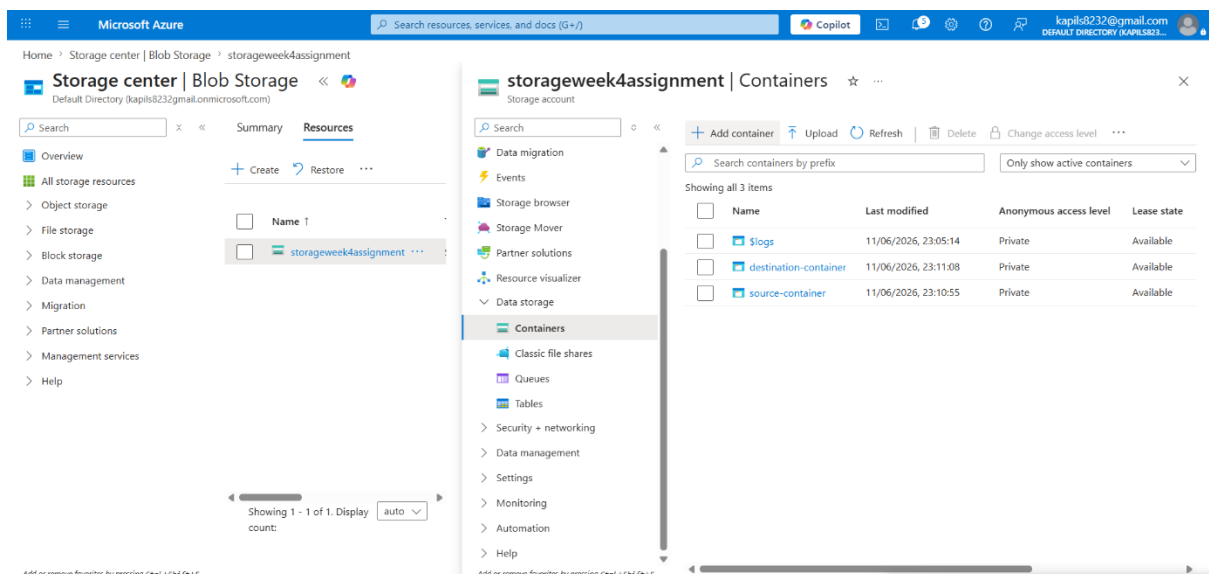
A Resource Group named (rg_week4) was created in Azure Portal to manage all resources used in the project.



Step 2: Storage Account and Blob Container

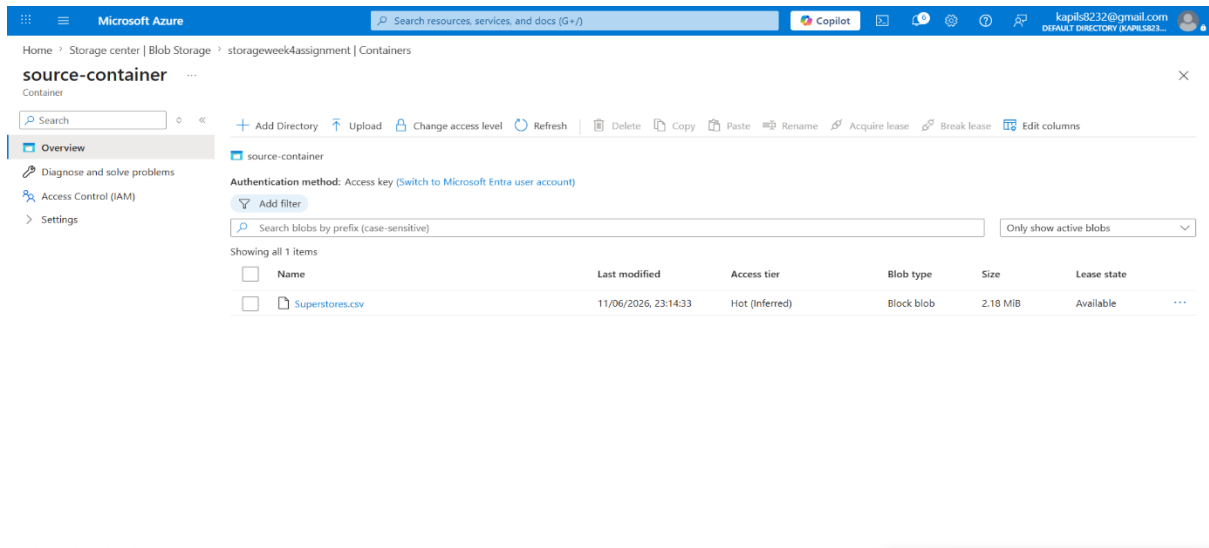
A Storage Account named (storageweek4assignment) and two Blob Containers are created.

- source-container
- destination container



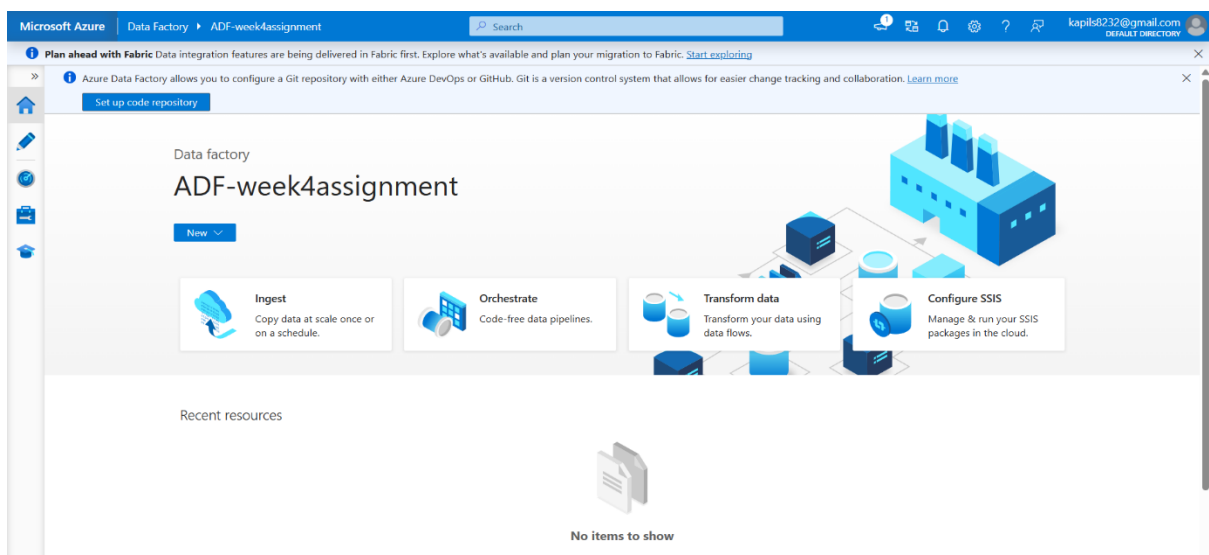
Step 3: Upload CSV File

A CSV file named **Superstore.csv** is uploaded to the source container in Azure Blob Storage. The file was successfully stored and verified in the container.



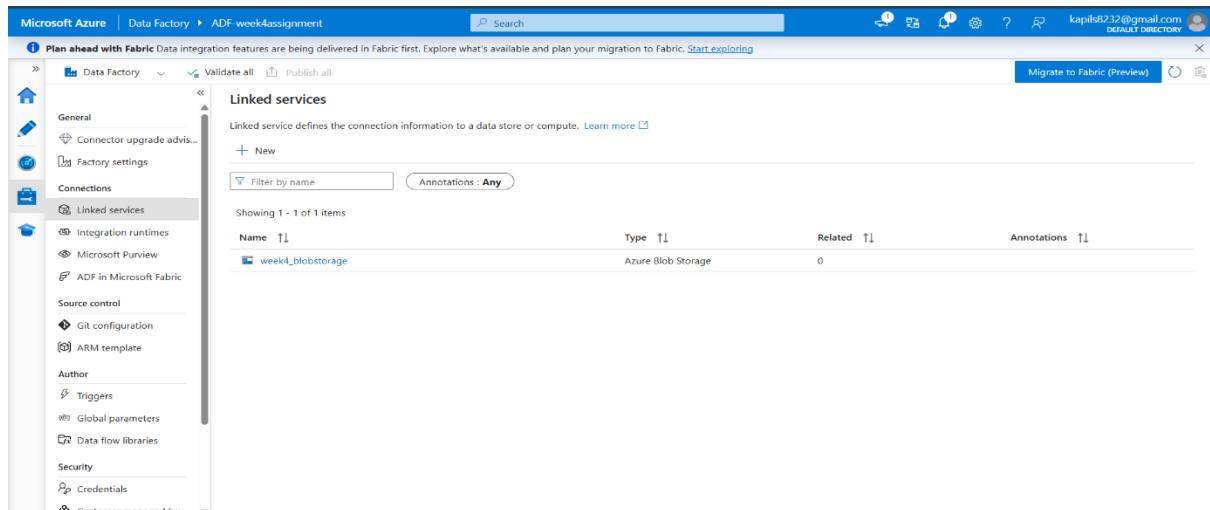
Step 4: Azure Data Factory Creation

Azure Data Factory is created named (ADF-week4assignment) and explored all the section of ADF (Author, Monitor, and Manage).



Step 5: Linked Service Creation

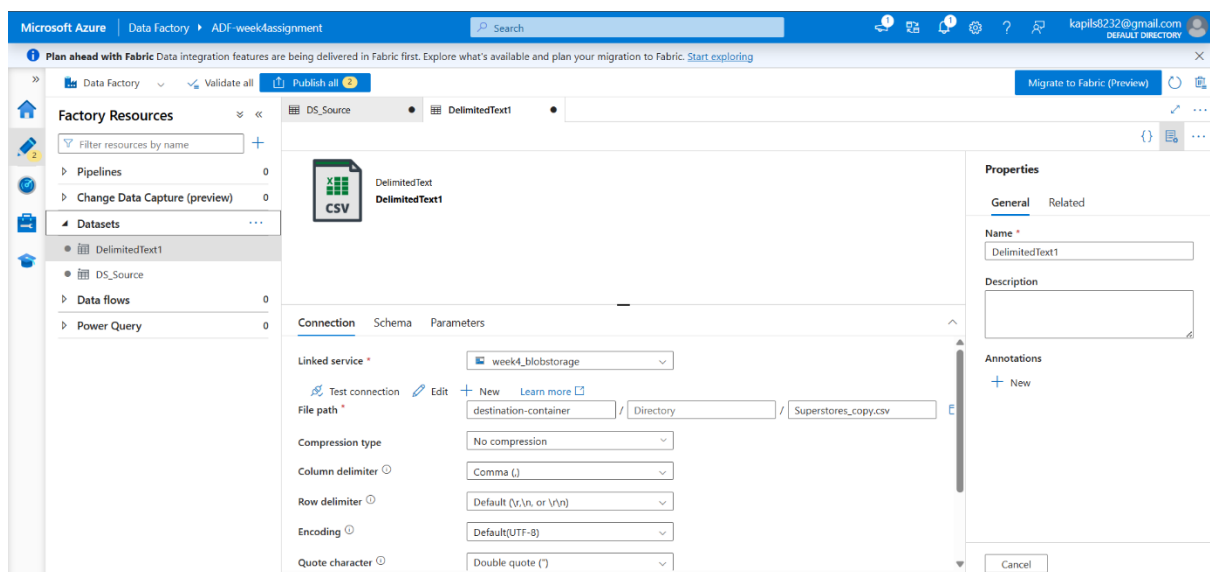
A Linked Service is created named (week4_blobstorage) to establish connection between Azure Data Factory and Azure Blob Storage



Step 6: Dataset Creation

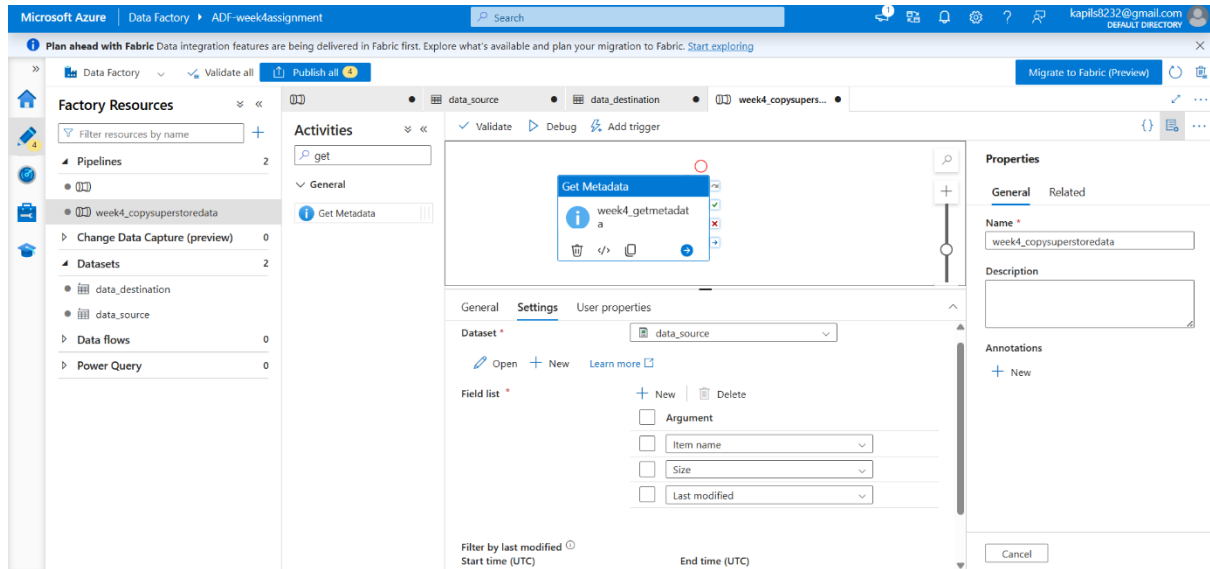
I set up two datasets - one for the source (the CSV file I uploaded) and one for the destination (where the copied file go)

- Source Dataset = DS_Source
- Destination Dataset = DelimitedText1



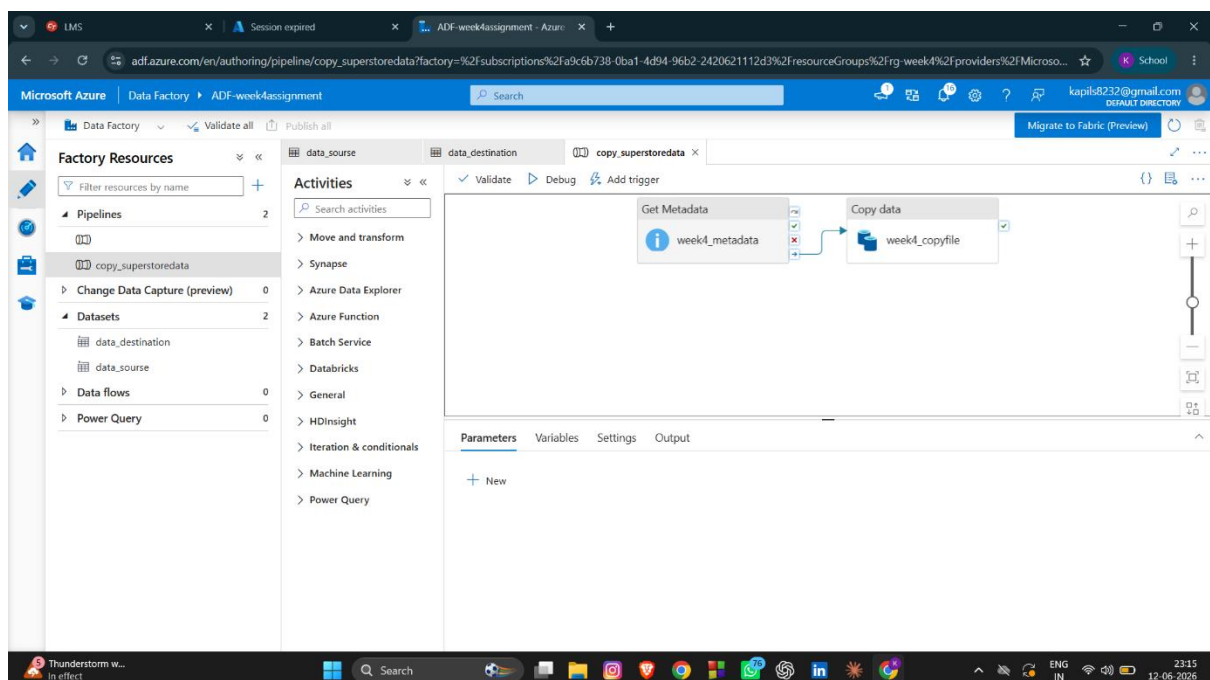
Step 7: Metadata Validation

Get Metadata activity is used to retrieve file information such as file name, file size, and last modified date just to make sure it's there and correct before moving it.



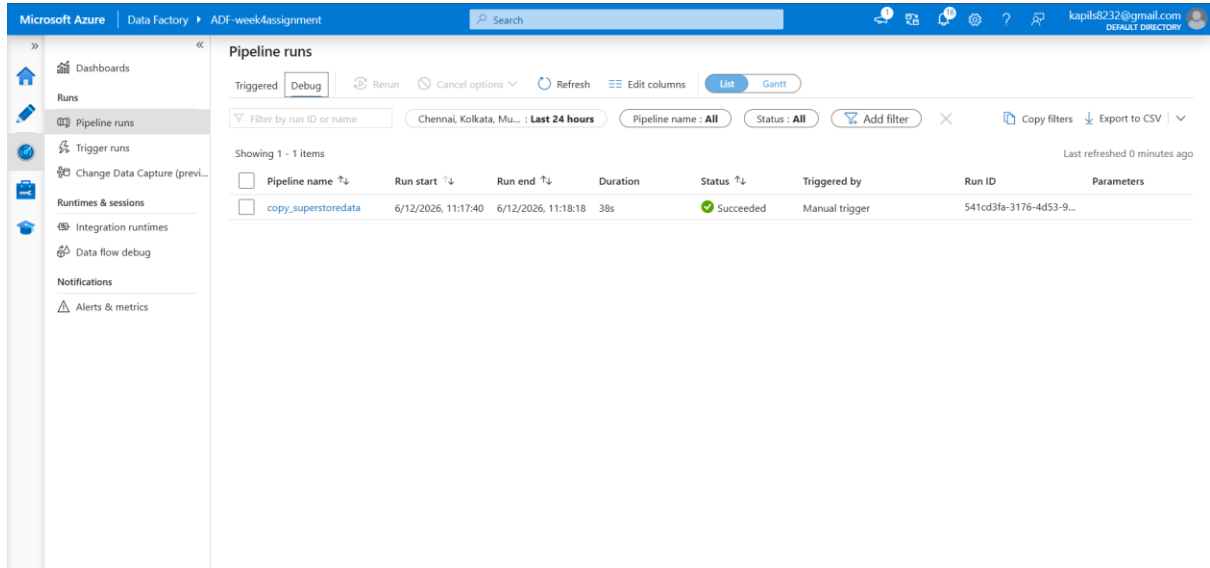
Step 8: Pipeline Development

I created the pipeline named (week4_copysuperstoredata) by adding a Copy Data activity after the Get Metadata step. This copies the file from the source location to the destination



Step 9: Pipeline Execution

The pipeline is executed using the Debug option in Azure Data Factory. The execution completed successfully, and the file was processed without any errors.

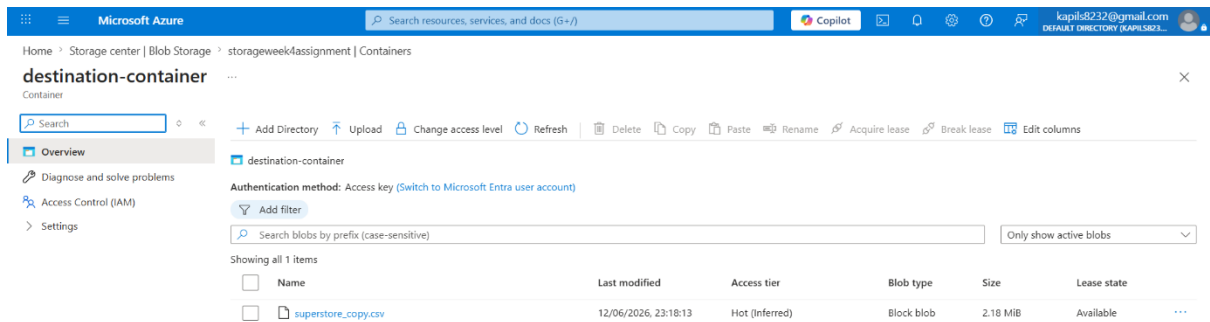


The screenshot shows the 'Pipeline runs' page in the Azure Data Factory portal. The left sidebar contains navigation options: Dashboards, Runs, Pipeline runs (selected), Trigger runs, Change Data Capture (previous), Runtimes & sessions, Integration runtimes, Data flow debug, Notifications, and Alerts & metrics. The main area displays the 'Pipeline runs' for the 'copy_superstoredata' pipeline. The 'Debug' tab is selected, and the 'Last 24 hours' filter is applied. The table shows one successful run.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
copy_superstoredata	6/12/2026, 11:17:40	6/12/2026, 11:18:18	38s	Succeeded	Manual trigger	541cd3fa-3176-4d53-9...	

Step 10: Output Validation

The output is validated by checking the destination container in Azure Blob Storage. The file was successfully copied from the source container and was available in the destination location.



The screenshot shows the 'destination-container' in the Azure Blob Storage portal. The left sidebar contains navigation options: Overview (selected), Diagnose and solve problems, Access Control (IAM), and Settings. The main area displays the 'destination-container' with the 'Access key' authentication method. The table shows one file, 'superstore_copy.csv'.

Name	Last modified	Access tier	Blob type	Size	Lease state
superstore_copy.csv	12/06/2026, 23:18:13	Hot (Inferred)	Block blob	2.18 MiB	Available

Step 11: IAM Role Assignment

Assigned Reader and Contributor roles to Data Factory. This gives ADF the permission it needs to read and write data in the storage account.

The screenshot displays the Microsoft Azure portal interface for the 'storageweek4assignment' storage account. The left-hand navigation pane includes links to Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage, Security + networking, Data management, Settings, Monitoring, and Automation. The 'Access Control (IAM)' section is currently selected.

The main content area shows the 'Role assignments' tab. At the top, there are buttons for '+ Add', 'Download role assignments', 'Edit columns', 'Refresh', 'Delete', and 'Feedback'. Below these, a summary indicates 'Number of role assignments for this subscription' with a count of 4 and a limit of 4000. A search bar labeled 'Search by name or email' is present, along with filters for 'Type: All', 'Role: All', 'Scope: All scopes', and 'Group by: Role'.

The role assignments are listed in a table with columns: Name, Type, Role, Scope, and Condition. The table shows four results, grouped into three categories:

- Owner (2)**: This group is currently collapsed.
- Reader (1)**: This group is expanded, showing one role assignment:
 - ADF-week4assignment** (Managed identity) assigned the **Reader** role to **This resource** with a condition of **None**.
- Storage Blob Data Contributor (1)**: This group is expanded, showing one role assignment:
 - ADF-week4assignment** (Managed identity) assigned the **Storage Blob Data Contributor** role to **This resource** with a condition of **Add**.

At the bottom of the table, it states 'Showing 1 - 4 of 4 results.'